

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409878
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Yamada; Tatsuya et al.

Steering device and method of wrapping synthetic leather

Abstract

The steering device includes a main body including a metal core, and a synthetic leather part wrapped around the main body. The synthetic leather part includes a first synthetic leather part, a second synthetic leather part made of a different material from the first synthetic leather part, and a first sewing part having a linear shape formed by sewing the first synthetic leather part and the second synthetic leather part together. The first sewing part is formed to extend in a direction not toward a center of the steering device in a front view of the steering device.

Inventors: Yamada; Tatsuya (Kiyosu, JP), Egawa; Seiji (Kiyosu, JP), Nemoto; Kazuki (Kiyosu, JP), Nakamura; Shintaro (Kiyosu, JP)

Applicant: TOYODA GOSEI CO., LTD. (Kiyosu, JP)

Family ID: 92899246

Assignee: TOYODA GOSEI CO., LTD. (Aichi-pref., JP)

Appl. No.: 18/602109

Filed: March 12, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240326893 A1	Oct. 03, 2024

Foreign Application Priority Data

JP	2023-053604	Mar. 29, 2023
JP	2024-026295	Feb. 26, 2024

Publication Classification

Int. Cl.: **B62D1/04** (20060101); **B32B1/00** (20240101); **B32B3/02** (20060101); **B32B3/14** (20060101); **B32B7/04** (20190101); **B32B7/09** (20190101); **B32B9/02** (20060101); **B32B15/04** (20060101); **B62D1/06** (20060101); B32B15/08 (20060101)

U.S. Cl.:

CPC **B62D1/06** (20130101); **B32B3/14** (20130101); **B32B7/09** (20190101); B32B1/00 (20130101); B32B9/025 (20130101); B32B15/08 (20130101); B32B2307/536 (20130101); B32B2307/546 (20130101); B32B2605/003 (20130101)

Field of Classification Search

CPC: B62D (1/06); B32B (1/00); B32B (3/14); B32B (7/09); B32B (9/025); B32B (15/08); B32B (2307/536); B32B (2307/546); B32B (2605/003)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4179950	12/1978	Valley	74/558	G05G 1/10
6065366	12/1999	Koyama	428/317.1	B62D 1/06
9598098	12/2016	Sakurai	N/A	B62D 1/04
9758187	12/2016	Nonoyama	N/A	B62D 1/065
9789895	12/2016	Yasui	N/A	B32B 9/025
10836420	12/2019	Lai	N/A	B32B 9/047
11066092	12/2020	Nonoyama	N/A	B60R 11/00
11390312	12/2021	Hotta	N/A	C09D 11/52
11584423	12/2022	Shinmen	N/A	B32B 5/265
11912332	12/2023	Quintard	N/A	B62D 1/06
2009/0178509	12/2008	Seidl	N/A	N/A
2014/0224059	12/2013	Morita	74/558	B62D 1/065

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1686743	12/2004	CN	N/A
218172379	12/2021	CN	N/A
115716499	12/2022	CN	B62D 1/06
2000108907	12/1999	JP	N/A
2009-165651	12/2008	JP	N/A
2014094728	12/2013	JP	N/A
2022067874	12/2021	JP	N/A
0132355	12/1997	KR	N/A
20160135963	12/2015	KR	N/A
M584764	12/2018	KR	N/A
WO-2020173736	12/2019	WO	B62D 1/06

OTHER PUBLICATIONS

Machine translation of KR 0132355 Y1 obtained on Apr. 24, 2025. cited by examiner

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The present application claims the priority to Japanese patent application No. 2023-053604 filed on Mar. 29, 2023, and Japanese patent application No. 2024-026295 filed on Feb. 26, 2024. The entire disclosure thereof is incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to a steering device and a synthetic leather wrapping method.

(3) The technique for wrapping leather around a grip part of a steering device is known well. Japanese patent application laid-open publication No. 2009-165651 discloses the technique of butting and sewing together two end portions of leather.

(4) When stress is applied, distortion may occur in the sewing part where the two end portions of leather are butted and sewed, in some cases. In the well-known technique for natural leather, its heat shrinking property is utilized to reduce the distortion in the sewing part.

(5) There is a demand of using synthetic leather as leather. Synthetic leather, which has lower heat resistance compared to natural leather, hardly allows heat treatment with application of heat to be used to reduce distortion in a sewing part.

SUMMARY

(6) The present disclosure has been made to solve the above-described problem, and may be embodied in the following aspects.

(7) (1) In one aspect of the present disclosure, a steering device is provided. The steering device includes a main body including a metal core, and a synthetic leather part wrapped around the main body. The synthetic leather part has a first synthetic leather part, a second synthetic leather part, and a first sewing part. The second synthetic leather part is made of a different material from the first synthetic leather part. The first sewing part has a linear shape formed by sewing the first synthetic leather part and the second synthetic leather part together. The first sewing part is formed to extend in a direction not toward a center of the steering device in a front view of the steering device.

(8) (2) In the steering device in the aspect above, the first synthetic leather part may have different hardness from the second synthetic leather part.

(9) (3) In the steering device in the aspect above, the first synthetic leather part may have different stretchability from the second synthetic leather part.

(10) (4) In the steering device in the aspect above, the main body may have an upper part, a grip part, and a lower part. The synthetic leather part may further have a third synthetic leather part and a second sewing part. The third synthetic part is made of a different material from the second synthetic leather part. The second sewing part has a linear shape formed by sewing the second synthetic leather part and the third synthetic leather part together. The first synthetic leather part may be wrapped around the upper part, and may be made of a harder material than the second synthetic leather part. The second synthetic leather part may be wrapped around the grip part, and may be made of a more stretchable material than the first synthetic leather part and the third synthetic leather part. The third synthetic leather part may be wrapped around the lower part.

(11) The present disclosure may be embodied in various aspects, for example, a synthetic leather member wrapped around a main body of a steering device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments of the disclosure are disclosed in the following detailed description and the accompanying drawings.
- (2) FIG. 1 is a schematic diagram illustrating an overview of the steering device,
- (3) FIG. 2 is an explanatory diagram illustrating sewing of a first synthetic leather part and a second synthetic leather part,
- (4) FIG. 3 is an explanatory diagram illustrating a comparison example of a first sewing part,
- (5) FIG. 4 is an explanatory diagram illustrating the first sewing part,
- (6) FIG. 5 is a schematic sectional diagram obtained by cutting at a V-V line in FIG. 1,
- (7) FIG. 6 is a flowchart indicating one example of a synthetic leather wrapping process.

DETAILED DESCRIPTION

A. First Embodiment

- (8) FIG. 1 is the schematic diagram illustrating the overview of a steering device **100** in the present embodiment. FIG. 1 shows the front view of the steering device **100**. The steering device **100** is held by a vehicle driver. The steering device **100** is operated by a driver, so that the vehicle is steered. The steering device **100** has a center oc as a steering rotation axis. In the present embodiment, the steering device **100** has a substantially rectangular shape. More specifically, the steering device **100** has a substantially trapezoidal shape in which its upper part is shorter than its lower part. The steering device **100** having such a shape is also called an atypical steering device.
- (9) The steering device **100** is configured with a main body **10** wrapped with a synthetic leather part **20**. In the present embodiment, the main body **10** includes a metal core **11** made of a metal member. The metal core **11**, for example, may have an annular shape or a substantially U-shaped. In the present embodiment, the metal core has a substantially U-shaped as shown in the cross-sectional view of FIG. 5. Since the metal core **11** has the substantially U-shaped, the weight of the metal core **11** is lighter than when the metal core **11** has an annular shape. The metal core **11** is arranged such that the substantially U-shaped opening side is directed toward the front of the vehicle, thereby breaking at the time of collision and absorbing collision energy. The main body **10** may include a cushion layer **12** covering the periphery of the metal core **11**. The cushion layer **12** is formed of, for example, a resin such as polyurethane, and can improve grip feeling and/or provide electrical insulation performance. The main body **10** may further include a functional layer **13** on the outside of the cushion layer **12**. The functional layer **13** is a layer that provides a desired functions, for example, a capacitive sensor sheet that provides a function of detecting gripping of the steering device **100**, and/or a heater element that provides a steering heater function. The main body **10** has an upper part, a lower part longer than the upper part, and grip parts each connecting the upper part and the lower part.
- (10) In the present embodiment, the synthetic leather part **20** has a first synthetic leather part **20A**, second synthetic leather parts **20B**, a third synthetic leather part **20C**, first sewing parts **S1**, and second sewing parts **S2**. The first sewing parts **S1** are the linear portions formed by sewing together the first synthetic leather part **20A** and the second synthetic leather parts **20B**, respectively. The second sewing parts **S2** are the linear portions formed by sewing together the second synthetic leather parts **20B** and the third synthetic leather part **20C**, respectively.
- (11) The steering device **100** has a first part **P1**, second parts **P2**, and a third part **P3**. The first part **P1** is configured with the main body **10** wrapped with the first synthetic leather part **20A**. In the present embodiment, the first synthetic leather part **20A** is wrapped around the upper part of the main body **10**. In other words, the upper part of the main body **10** corresponds to the first part **P1**. The second parts **P2** are configured with the main body **10** wrapped with the second synthetic leather parts **20B**, respectively. In the present embodiment, the second synthetic leather parts **20B**

are wrapped around the grip parts of the main body **10**, respectively. In other words, the grip parts of the main body **10** correspond to the second parts **P2**. The third part **P3** is configured with the main body **10** wrapped with the third synthetic leather part **20C**. In the present embodiment, the third synthetic leather part **20C** is wrapped around the lower part of the main body **10**. In other words, the lower part of the main body **10** corresponds to the third part **P3**.

(12) The boundary parts between the first part **P1** and the second parts **P2** correspond to the first sewing parts **S1**, respectively. The boundary parts between the second parts **P2** and the third part **P3** correspond to the second sewing parts **S2**, respectively. In the present embodiment not shown in the drawings, in the first part **P1** and the third part **P3**, the synthetic part **20** may be wrapped only at a part of the area near the end adjoining the second part **P2**.

(13) As shown in FIG. 1, in the front view of the steering device **100**, each of the first sewing parts **S1** is formed to extend in a direction not toward the center oc. In addition, in the front view of the steering device **100**, each of the second sewing parts **S2** is formed to extend in a direction not toward the center oc. The extending directions of the first sewing parts **S1** and the second sewing parts **S2** are set on the basis of the shape of the main body **10** and/or the design of the steering device **100**.

(14) The first synthetic leather part **20A** is made of a different material from the second synthetic leather parts **20B**. The third synthetic leather part **20C** is made of a different material from the second synthetic leather parts **20B**. In the present embodiment, the first synthetic leather part **20A** is harder than the second synthetic leather parts **20B**. The third synthetic leather part **20C** is harder than the second synthetic leather parts **20B**. In the present disclosure, “a different material” refers to a material different in at least one of hardness and stretchability. For example, a material just different in color is not treated as “a different material”.

(15) In the present embodiment, the hardness of a material is represented by the depression amount measured by ST300 Softness Tester of MSA Engineering Systems Limited, with a ring diameter set to 35 mm. As a material is softer, the depression amount of the material is larger. The depression amount of the first synthetic leather part **20A** is 4.4 mm, and the depression amount of the second synthetic leather parts **20B** is 5.0 mm. The depression amount of the third synthetic leather part **20C** is 3.4 mm. In other words, the first synthetic leather part **20A** is harder than the second synthetic leather parts **20B**, and the third synthetic leather part **20C** is harder than the first synthetic leather part **20A**.

(16) The second synthetic leather parts **20B** are made of a more stretchable material than the first synthetic leather part **20A** and the third synthetic leather part **20C**. In the present embodiment, the stretchability of a material is represented by the load at which the material is deformed, measured by Technograph TG-10KN Tensile Compression Testing Machine by MinebeaMitsumi Inc., with a crosshead speed set at 200 mm/sec. As a material is more stretchable, the load value of the material is smaller. The load of the first synthetic leather part **20A** is 133 N, the load of the second synthetic leather parts **20B** is 109 N, and the load of the third synthetic leather part **20C** is 181 N. In other words, the second synthetic leather parts **20B** are more stretchable than the first synthetic leather part **20A**, and the first synthetic leather part **20A** is more stretchable than the third synthetic leather part **20C**.

(17) FIG. 2 is the explanatory diagram illustrating the sewing of the first synthetic leather part **20A** and the second synthetic leather part **20B**. Since the first sewing part **S1** is formed to extend in a direction not toward the center oc, the length of the first synthetic leather part **20A** along the sewing direction of the first sewing part **S1** is shorter than the length of the second synthetic leather part **20B** along the sewing direction of the first sewing part **S1**. In the present embodiment, the first synthetic leather part **20A** is sewed to the second synthetic leather part **20B** while the edge part of the first synthetic leather part **20A** is pulled in the direction of the arrows shown in FIG. 2.

Accordingly, stress is applied to the synthetic leather part **20** in the opposite direction to the arrows shown in FIG. 2.

(18) FIG. 3 is the explanatory diagram illustrating a first sewing part **S1a** as a comparison example. A synthetic leather part **30** of the comparison example shown in FIG. 3 is different from the synthetic leather part **20** according to the present embodiment in that a first synthetic leather part **30A** is made of the same material as a second synthetic leather part **30B**. Since the first synthetic leather part **30A** is made of the same material as the second synthetic leather part **30B**, the stress which has been applied to the synthetic leather part **20** but has not been cancelled out by either the first synthetic leather part **30A** or the second synthetic leather part **30B** is applied to the first sewing part **S1a**. This causes significant distortion in the first sewing part **S1a**, as shown in FIG. 3.

(19) FIG. 4 is the explanatory diagram illustrating the first sewing part **S1**. As shown in FIG. 4, in the present embodiment, since the first synthetic leather part **20A** is made of a different material from the second synthetic leather part **20B**, the stress which has been applied to the synthetic leather part **20** but has not been cancelled out by either the first synthetic leather part **20A** or the second synthetic leather part **20B** is applied to, for example, the first synthetic leather part **20A**. This reduces the occurrence of distortion in the first sewing part **S1**, as shown in FIG. 4.

(20) FIG. 5 is the schematic sectional diagram obtained by cutting the steering device **100** at the V-V line in FIG. 1. As shown in FIG. 5, the synthetic leather part **20** is wrapped outer surface of the main body **10** including the metal core **11**, the cushion layer **12** and the functional layer **13**.

(21) In the steering device **100** according to the present embodiment described above, the first synthetic leather part **20A** is made of a different material from the second synthetic leather parts **20B**. Since each of the first sewing parts **S1** is formed to extend in a direction not toward the center **oc** in the front view of the steering device **100**, the stress which has been applied to the synthetic leather part **20** but has not been cancelled out by, for example, the second synthetic leather parts **20B** is able to be cancelled out by the first synthetic leather part **20A** made of a harder material than the second synthetic leather parts **20B**. This reduces the occurrence of distortion in the first sewing parts **S1**, and accordingly reduces the occurrence of distortion in the synthetic leather part **20**.

(22) The first synthetic leather part **20A** has a coating layer **21**, a base fabric layer **22**, and a lining layer **23**. The first synthetic leather part **20A** is formed by laminating the coating layer **21**, the base fabric layer **22**, and the lining layer **23**, in this order. The synthetic leather part **20** covers the main body **10** so that the coating layer **21** is arranged as the surface. The coating layer **21** is made of, for example, resin. The base fabric layer **22** is made of, for example, woven fabric, knitted fabric, or non-woven fabric. The lining layer **23** is made of, for example, sponge such as urethane foam. As with the first synthetic leather part **20A**, the second synthetic leather parts **20B** and the third synthetic leather part **20C**, respectively, have laminated layers of the coating layer **21**, the base fabric layer **22**, and the lining layer **23**.

(23) FIG. 6 is a flowchart indicating one example of a synthetic leather wrapping process. The synthetic leather wrapping process is executed to wrap the synthetic leather part **20** around the main body **10**. Step **S100** involves preparing the main body **10**. The step is also referred to as “preparation step”.

(24) Step **S110** involves sewing together the first synthetic leather part **20A** and the second synthetic leather parts **20B** respectively, and sewing together the second synthetic leather parts **20B** and the third synthetic leather part **20C** respectively. In the present embodiment, the first synthetic leather part **20A** and the second synthetic leather part **20B** are sewed together while the first synthetic leather part **20A** is pulled in the width direction of the first synthetic leather part **20A**. The first synthetic leather part **20A** is made of a harder material than the second synthetic leather parts **20B**, thereby reducing the occurrence of distortion in the first sewing parts **S1**. The second synthetic leather parts **20B** and the third synthetic leather part **20C** are sewed together while the third synthetic leather part **20C** is pulled in the width direction of the third synthetic leather part **20C**. The third synthetic leather part **20C** is made of a less stretchable material than the second synthetic leather parts **20B**, thereby reducing the occurrence of distortion in the second sewing parts **S2**. The step is also referred to as “sewing step”.

(25) Step **S120** involves wrapping the synthetic leather part **20** around the main body **10**. More specifically, the step involves wrapping the first synthetic leather part **20A** around the upper part of the main body **10**, wrapping the second synthetic leather parts **20B** around the grip parts of the main body **10**, and wrapping the third synthetic leather part **20C** around the lower part of the main body **10**. The step is also referred to as “wrapping step”. Step **S110** and step **S120** may be executed in any order, not limited to this order, or may be executed in parallel. The combination of step **S110** and step **S120** is also referred to as “synthetic leather part forming step”.

(26) In the steering **100** according to the present embodiment described above, the first synthetic leather part **20A** is made of a different material from the second synthetic leather parts **20B**. Since each of the first sewing parts **S1** is formed to extend in a direction not toward the center oc in the front view of the steering **100**, the stress which has been applied to the synthetic leather part **20** but has not been cancelled out by, for example, the second synthetic leather parts **20B** is able to be cancelled out by the first synthetic leather part **20A** made of a harder material than the second synthetic leather parts **20B**. This reduces the occurrence of distortion in the first sewing parts **S1**, and accordingly reduces the occurrence of distortion in the synthetic leather part **20**.

(27) The second synthetic leather parts **20B** are made of a different material from the third synthetic leather part **20C**. Since each of the second sewing parts **S2** is formed to extend in a direction not toward the center oc in the front view of the steering device **100**, the stress which has been applied to the synthetic leather part **20** but has not been cancelled out by, for example, the second synthetic leather parts **20B** is able to be cancelled out by the third synthetic leather part **20C** made of a less stretchable material than the second synthetic leather parts **20B**. This reduces the occurrence of distortion in the second sewing parts **S2**, and accordingly reduces the occurrence of distortion in the synthetic leather part **20**.

(28) The second synthetic leather parts **20B** are made of a more stretchable material than the first synthetic leather part **20A** and the third synthetic leather part **20C**. Therefore, when the second parts **P2** has a larger area than the area of the first part **P1** and the area of the third part **P3**, by covering the second part **20B** with the second synthetic leather having a smaller area than the second part **P2** area, appearance of wrinkles in the synthetic leather part **20** can be reduced.

B. Other Embodiments

(29) (B1) In the embodiment described above, the steering device **100** has a substantially rectangular shape. Alternatively, the steering device **100** may have a substantially circular shape or a substantially elliptical shape. Further, the steering device **100** may have a control-stick-like shape, or a letter-D shape where a substantially elliptical shape partially has a straight line.

(30) (B2) In the embodiment described above, the synthetic leather part **20** has the first synthetic leather part **20A**, the second synthetic leather parts **20B**, and the third synthetic leather part **20C**. Alternatively, the synthetic leather part **20** may have at least the first synthetic leather part **20A** and the second synthetic leather part **20B**. For example, the synthetic leather part **20** wrapped around the main body **10** in the third part **P3** may be the first synthetic leather part **20A**.

(31) (B3) In the embodiment described above, the first synthetic leather part **20A** has different hardness and stretchability from the second synthetic leather parts **20B**. In another example, as long as the first synthetic leather part **20A** is made of a different material from the second synthetic leather parts **20B**, the first synthetic leather part **20A** may have different hardness from the second synthetic leather parts **20B**, and the same stretchability as the second synthetic leather parts **20B**.

(32) (B4) In the embodiment described above, the third synthetic leather part **20C** is harder than the first synthetic leather part **20A**. Alternatively, the third synthetic leather part **20C** may have the same hardness as or lower hardness than the first synthetic leather part **20A**. The first synthetic leather part **20A** is more stretchable than the third synthetic leather part **20C**. Alternatively, the first synthetic leather part **20A** may have the same stretchability as or lower stretchability than the third synthetic leather part **20C**.

(33) The present disclosure may be embodied in various configurations without departing from the

spirit of the present disclosure, not limited to the above-described embodiments. For example, the technical features in the embodiments corresponding to the technical features in the respective aspects disclosed in the summary above may be appropriately replaced or combined in order to solve the above-described problem, or in order to achieve some or all of the above-described effects. Any of the technical features not described as essential in the present specifications may be omitted appropriately.

Claims

1. A steering device comprising: a main body including a metal core; and a synthetic leather part wrapped around the main body, wherein the synthetic leather part comprises a first synthetic leather part, a second synthetic leather part, and a first sewing part, the second synthetic leather part is made of a different material from the first synthetic leather part, the first sewing part has a linear shape formed by sewing the first synthetic leather part and the second synthetic leather part together, and the first sewing part is formed to extend in a direction not toward a center of the steering device in a front view of the steering device, wherein the first synthetic leather part has a different hardness from the second synthetic leather part.
 2. A steering device comprising: a main body including a metal core; and a synthetic leather part wrapped around the main body, wherein the synthetic leather part comprises a first synthetic leather part, a second synthetic leather part, and a first sewing part, the second synthetic leather part is made of a different material from the first synthetic leather part, the first sewing part has a linear shape formed by sewing the first synthetic leather part and the second synthetic leather part together, and the first sewing part is formed to extend in a direction not toward a center of the steering device in a front view of the steering device, wherein the first synthetic leather part has a different stretchability from the second synthetic leather part.
 3. A steering device comprising: a main body including a metal core; and a synthetic leather part wrapped around the main body, wherein the synthetic leather part comprises a first synthetic leather part, a second synthetic leather part, and a first sewing part, the second synthetic leather part is made of a different material from the first synthetic leather part, the first sewing part has a linear shape formed by sewing the first synthetic leather part and the second synthetic leather part together, and the first sewing part is formed to extend in a direction not toward a center of the steering device in a front view of the steering device, wherein the main body comprises an upper part, a grip part, and a lower part, the synthetic leather part further comprises a third synthetic leather part and a second sewing part, the third synthetic leather part is made of a different material from the second synthetic leather part, and the second sewing part has a linear shape formed by sewing the second synthetic leather part and the third synthetic leather part together, the first synthetic leather part is wrapped around the upper part, and is made of a harder material than the second synthetic leather part, the second synthetic leather part is wrapped around the grip part, and is more stretchable than the first synthetic leather part and the third synthetic leather part, and the third synthetic leather part is wrapped around the lower part.
-